# RIG-ETALE PERMANENCE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-rig-etale-permanence; the proof environment immediately following it is its own proof

*Editorial note.* The label lemma-rig-etale-permanence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- the WAdm^Noeth-level permanence lemma for rig-etale arrows (lemma-permanence-rig-etale-continuous of the same chapter)
- the Formal Spaces Noetherian variant remark for permanence (formal-spaces-remark-permanence-variant-Noetherian)

*Source reference.* Original remark 0GBP: sources/stacks-chow/tags.txt, source lines 20379–20379. Prerequisite pointer invoked at 5162; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0GCY: sources/stacks-restricted/pinned-original-source.tex, source lines 4965–4965. Prerequisite pointer invoked at 5163; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be a morphism of locally Noetherian formal algebraic spaces over $S$. If $g \circ f$ and $g$ are rig-étale, then so is $f$.*

*Proof.* By Formal Spaces, Remark 0GBP this follows from Lemma 0GCY. □

*Date*: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.